POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6042 | Semester VI | Theory | 4 Credits

Robotics and Artificial Intelligence

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6042 as Robotics and Artificial Intelligence in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Introduction to Robotics course and polytechnic AI-robotics curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Robotic designs, AI models, kinematic analysis and autonomous systems must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support robot kinematics, AI algorithms, sensor integration, and path planning. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Robotics and Artificial Intelligence studies the design, control and intelligent behavior of robotic systems. It develops the ability to analyze robot kinematics, evaluate AI algorithms for perception and decision-making, understand the integration of smart sensors and actuators, and coordinate autonomous robotic solutions while respecting the technical and safety boundaries of electronic and robotic engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Digital electronics, control systems, basic programming and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamental components, classification and kinematics of robotic systems.
2. Apply artificial intelligence principles including search and machine learning to robotic tasks.
3. Analyze the interfacing of intelligent sensors and actuators for robotic perception and motion.
4. Explain the systematic design and path planning process for autonomous robots in various applications.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the fundamentals of robotics including robot components and classification.	Understand
CO2	Apply AI search and learning algorithms to solve robotic problems.	Apply
CO3	Analyze the kinematics and control of robotic manipulators.	Apply
CO4	Design path planning and perception systems for autonomous robots.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Robotics Fundamentals and Components	Definition and history of robotics; Laws of robotics; Robot classification: Industrial, Mobile, Humanoid; Robot components: Links, Joints, End effectors; Sensors (Lidar, Ultrasonic, Vision); Actuators (Servos, Stepper motors).	Approx. 14 hours	CO1	Module 1
2	Robot Kinematics and Control	Coordinate frames and transformations; Homogeneous transformation matrices; DH parameters; Forward and Inverse kinematics (2-DOF/3-DOF); Trajectory planning; PID control for joint motion.	Approx. 16 hours	CO2	Module 2
3	Artificial Intelligence for Robotics	Introduction to AI; Problem solving using search: BFS, DFS, A* algorithm; Expert systems; Machine Learning basics: Supervised, Unsupervised, Reinforcement learning; Neural networks and Deep learning overview.	Approx. 16 hours	CO3	Module 3
4	Autonomous Systems and Applications	Path planning in dynamic environments; Obstacle avoidance; SLAM (Simultaneous Localization and Mapping) basics; Human-Robot Interaction (HRI); Industrial and service applications; Ethical issues in AI and robotics.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Robotics Fundamentals and Components

Definition and history of robotics; Laws of robotics; Robot classification: Industrial, Mobile, Humanoid; Robot components: Links, Joints, End effectors; Sensors (Lidar, Ultrasonic, Vision); Actuators (Servos, Stepper motors).

CO1 · Approx. 14 hours

Robot Kinematics and Control

Coordinate frames and transformations; Homogeneous transformation matrices; DH parameters; Forward and Inverse kinematics (2-DOF/3-DOF); Trajectory planning; PID control for joint motion.

CO2 · Approx. 16 hours

Artificial Intelligence for Robotics

Introduction to AI; Problem solving using search: BFS, DFS, A* algorithm; Expert systems; Machine Learning basics: Supervised, Unsupervised, Reinforcement learning; Neural networks and Deep learning overview.

CO3 · Approx. 16 hours

Autonomous Systems and Applications

Path planning in dynamic environments; Obstacle avoidance; SLAM (Simultaneous Localization and Mapping) basics; Human-Robot Interaction (HRI); Industrial and service applications; Ethical issues in AI and robotics.

CO4 · Approx. 14 hours

MODULE 1

Robotics Fundamentals and Components

Definition and history of robotics; Laws of robotics; Robot classification: Industrial, Mobile, Humanoid; Robot components: Links, Joints, End effectors; Sensors (Lidar, Ultrasonic, Vision); Actuators (Servos, Stepper motors).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

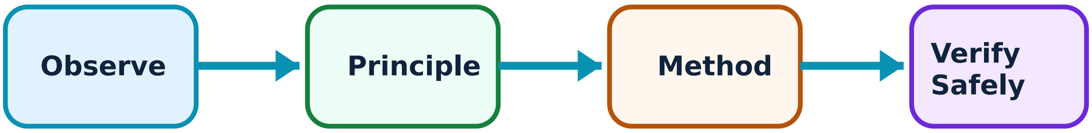
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Robotics Fundamentals and Components: identify the system, state its principle, follow the correct method and verify the result safely.

1. Introduction to Robotics−

Robotics is an interdisciplinary field involving the design, construction, and operation of robots. Robots are classified based on their workspace (Cartesian, Cylindrical, Spherical, SCARA) and application. The three laws of robotics provide an ethical framework for robot behavior.

Study and application method

State the 'Three Laws of Robotics'; compare 'SCARA' and 'Articulated' robots in a conceptual table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State the 'Three Laws of Robotics'; compare 'SCARA' and 'Articulated' robots in a conceptual table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Robots involve moving parts and high-torque motors. Do not operate industrial robots without authorised safety cages, emergency stops, and proximity sensors.

2. Robotic Sensors and Actuators−

Sensors allow robots to perceive their environment. Lidar and Ultrasonic sensors are used for distance measurement, while Vision sensors (cameras) enable object recognition. Actuators like Servo and Stepper motors provide precise motion control for joints and wheels.

Study and application method

Describe the role of 'Lidar' in autonomous navigation; explain how a 'Servo Motor' differs from a standard DC motor in robotics.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of 'Lidar' in autonomous navigation; explain how a 'Servo Motor' differs from a standard DC motor in robotics.

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Common mistake / precaution: Actuators can generate significant heat and inductive noise. Do not interface high-power motors without authorised drivers, isolation, and thermal management.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Robot Kinematics and Control

Coordinate frames and transformations; Homogeneous transformation matrices; DH parameters; Forward and Inverse kinematics (2-DOF/3-DOF); Trajectory planning; PID control for joint motion.

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

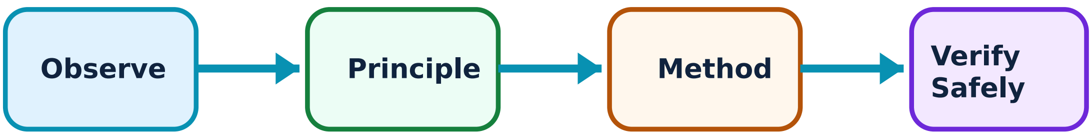
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Robot Kinematics and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Kinematic Modeling−

Kinematics studies the motion of robots without considering forces. Forward kinematics calculates the end-effector position from joint angles, while Inverse kinematics finds the required joint angles for a target position. Denavit-Hartenberg (DH) parameters are used to systematically model robot links.

Study and application method

Sketch a '2-DOF Planar Robot' and derive its forward kinematics equations; define 'Inverse Kinematics' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a '2-DOF Planar Robot' and derive its forward kinematics equations; define 'Inverse Kinematics' conceptually.

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Common mistake / precaution: Inverse kinematics solutions may be non-existent or multiple (singularities). All motion control software must follow authorised singularity-handling and workspace-limit algorithms.

2. Motion Control and Planning−

Trajectory planning generates a smooth path for the robot to follow. PID (Proportional-Integral-Derivative) controllers are used to minimize the error between the desired and actual joint positions. Advanced control involves compensating for gravity and inertia.

Study and application method

Explain the importance of 'Trajectory Planning' in robotic painting or welding; describe the role of the 'Derivative' term in a PID controller.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the importance of 'Trajectory Planning' in robotic painting or welding; describe the role of the 'Derivative' term in a PID controller.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Abrupt motion can damage the robot structure or payload. Do not implement high-speed trajectories without authorised acceleration limiting and jerk analysis.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Artificial Intelligence for Robotics

Introduction to AI; Problem solving using search: BFS, DFS, A* algorithm; Expert systems; Machine Learning basics: Supervised, Unsupervised, Reinforcement learning; Neural networks and Deep learning overview.

Approx. 16 hours **CO3** **Understand · Apply**

Learning goals

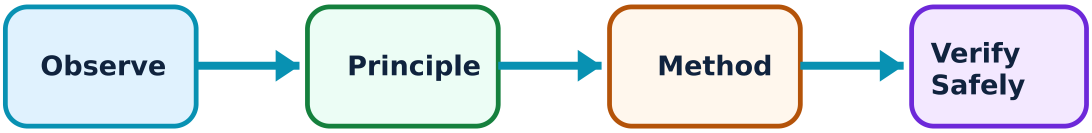
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Artificial Intelligence for Robotics: identify the system, state its principle, follow the correct method and verify the result safely.

1. AI Search and Problem Solving−

AI enables robots to make decisions. Search algorithms like A* are used for finding the shortest path in a known map. Expert systems use a set of rules to solve complex problems in specific domains like diagnostics.

Study and application method

Compare 'Breadth-First Search' (BFS) and 'Depth-First Search' (DFS) conceptually; explain the advantage of the 'A*' algorithm for path planning.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Breadth-First Search' (BFS) and 'Depth-First Search' (DFS) conceptually; explain the advantage of the 'A*' algorithm for path planning.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Search algorithms can be computationally expensive. Do not use unoptimized search on resource-constrained embedded platforms without authorised memory and CPU analysis.

2. Machine Learning and Perception−

Machine learning allows robots to improve their performance from experience. Supervised learning is used for object recognition, while Reinforcement learning is used for training robots to perform complex tasks through trial and error. Neural networks mimic the human brain to process sensory data.

Study and application method

Differentiate between 'Supervised' and 'Unsupervised' learning with examples; describe the role of 'Deep Learning' in computer vision for robots.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Supervised' and 'Unsupervised' learning with examples; describe the role of 'Deep Learning' in computer vision for robots.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: AI models are black boxes and can exhibit unpredictable behavior. Do not deploy AI-driven control in safety-critical tasks without authorised fail-safe overrides and human supervision.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Autonomous Systems and Applications

Path planning in dynamic environments; Obstacle avoidance; SLAM (Simultaneous Localization and Mapping) basics; Human-Robot Interaction (HRI); Industrial and service applications; Ethical issues in AI and robotics.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

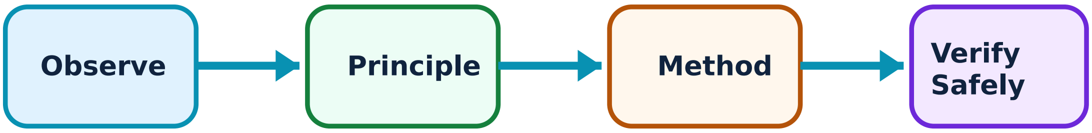
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Autonomous Systems and Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. Navigation and SLAM−

Autonomous robots must navigate without human intervention. Obstacle avoidance uses sensor data to modify the path in real-time. SLAM allows a robot to build a map of an unknown environment while keeping track of its own location within it.

Study and application method

Explain the concept of 'SLAM' (Simultaneous Localization and Mapping); describe a simple 'Obstacle Avoidance' algorithm.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the concept of 'SLAM' (Simultaneous Localization and Mapping); describe a simple 'Obstacle Avoidance' algorithm.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Localization errors can lead to collisions. All autonomous navigation systems must include authorised redundant sensors and physical bumpers.

2. Applications and Ethics−

Robotics and AI are transforming industries. Industrial robots perform repetitive tasks with high precision. Service robots assist in healthcare, cleaning, and delivery. Ethical considerations include job displacement, bias in AI algorithms, and the safety of autonomous weapons.

Study and application method

List three applications of 'Robotics in Healthcare'; discuss the 'Ethical Implications' of using autonomous vehicles on public roads.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List three applications of 'Robotics in Healthcare'; discuss the 'Ethical Implications' of using autonomous vehicles on public roads.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Ethical and legal standards for AI are evolving. All robotic deployments must follow authorised privacy, safety, and accountability guidelines as per local and international regulations.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Robotics Fundamentals and Components

Use the concept flow to revise observation, principle, method and verification.

Module 2: Robot Kinematics and Control

Use the concept flow to revise observation, principle, method and verification.

Module 3: Artificial Intelligence for Robotics

Use the concept flow to revise observation, principle, method and verification.

Module 4: Autonomous Systems and Applications

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Industrial and Mobile robots in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the state diagram for a 'Line Follower Robot' using two sensors.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the DH parameter table for a 2-DOF planar manipulator conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Apply the A* algorithm conceptually to find a path in a simple 5x5 grid with obstacles.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a robotic system in a local industry or a video demonstration (e.g., Amazon warehouse) and note its features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Robotics Fundamentals and Components

Explain Introduction to Robotics and state one correct method, application or precaution.−

Robotics is an interdisciplinary field involving the design, construction, and operation of robots. Robots are classified based on their workspace (Cartesian, Cylindrical, Spherical, SCARA) and application. The three laws of robotics provide an ethical framework for robot behavior.

Answer framework: State the 'Three Laws of Robotics'; compare 'SCARA' and 'Articulated' robots in a conceptual table.

Important point: Robots involve moving parts and high-torque motors. Do not operate industrial robots without authorised safety cages, emergency stops, and proximity sensors.

Explain Robotic Sensors and Actuators and state one correct method, application or precaution.−

Sensors allow robots to perceive their environment. Lidar and Ultrasonic sensors are used for distance measurement, while Vision sensors (cameras) enable object recognition. Actuators like Servo and Stepper motors provide precise motion control for joints and wheels.

Answer framework: Describe the role of 'Lidar' in autonomous navigation; explain how a 'Servo Motor' differs from a standard DC motor in robotics.

Important point: Actuators can generate significant heat and inductive noise. Do not interface high-power motors without authorised drivers, isolation, and thermal management.

Module 2 — Robot Kinematics and Control

Explain Kinematic Modeling and state one correct method, application or precaution. −

Kinematics studies the motion of robots without considering forces. Forward kinematics calculates the end-effector position from joint angles, while Inverse kinematics finds the required joint angles for a target position. Denavit-Hartenberg (DH) parameters are used to systematically model robot links.

Answer framework: Sketch a '2-DOF Planar Robot' and derive its forward kinematics equations; define 'Inverse Kinematics' conceptually.

Important point: Inverse kinematics solutions may be non-existent or multiple (singularities). All motion control software must follow authorised singularity-handling and workspace-limit algorithms.

Explain Motion Control and Planning and state one correct method, application or precaution.−

Trajectory planning generates a smooth path for the robot to follow. PID (Proportional-Integral-Derivative) controllers are used to minimize the error between the desired and actual joint positions. Advanced control involves compensating for gravity and inertia.

Answer framework: Explain the importance of 'Trajectory Planning' in robotic painting or welding; describe the role of the 'Derivative' term in a PID controller.

Important point: Abrupt motion can damage the robot structure or payload. Do not implement high-speed trajectories without authorised acceleration limiting and jerk analysis.

Module 3 — Artificial Intelligence for Robotics

Explain AI Search and Problem Solving and state one correct method, application or precaution.−

AI enables robots to make decisions. Search algorithms like A* are used for finding the shortest path in a known map. Expert systems use a set of rules to solve complex problems in specific domains like diagnostics.

Answer framework: Compare 'Breadth-First Search' (BFS) and 'Depth-First Search' (DFS) conceptually; explain the advantage of the 'A*' algorithm for path planning.

Important point: Search algorithms can be computationally expensive. Do not use unoptimized search on resource-constrained embedded platforms without authorised memory and CPU analysis.

Explain Machine Learning and Perception and state one correct method, application or precaution.−

Machine learning allows robots to improve their performance from experience. Supervised learning is used for object recognition, while Reinforcement learning is used for training robots to perform complex tasks through trial and error. Neural networks mimic the human brain to process sensory data.

Answer framework: Differentiate between 'Supervised' and 'Unsupervised' learning with examples; describe the role of 'Deep Learning' in computer vision for robots.

Important point: AI models are black boxes and can exhibit unpredictable behavior. Do not deploy AI-driven control in safety-critical tasks without authorised fail-safe overrides and human supervision.

Module 4 — Autonomous Systems and Applications

Explain Navigation and SLAM and state one correct method, application or precaution.–

Autonomous robots must navigate without human intervention. Obstacle avoidance uses sensor data to modify the path in real-time. SLAM allows a robot to build a map of an unknown environment while keeping track of its own location within it.

Answer framework: Explain the concept of 'SLAM' (Simultaneous Localization and Mapping); describe a simple 'Obstacle Avoidance' algorithm.

Important point: Localization errors can lead to collisions. All autonomous navigation systems must include authorised redundant sensors and physical bumpers.

Explain Applications and Ethics and state one correct method, application or precaution.–

Robotics and AI are transforming industries. Industrial robots perform repetitive tasks with high precision. Service robots assist in healthcare, cleaning, and delivery. Ethical considerations include job displacement, bias in AI algorithms, and the safety of autonomous weapons.

Answer framework: List three applications of 'Robotics in Healthcare'; discuss the 'Ethical Implications' of using autonomous vehicles on public roads.

Important point: Ethical and legal standards for AI are evolving. All robotic deployments must follow authorised privacy, safety, and accountability guidelines as per local and international regulations.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Robotics Fundamentals and Components.

2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Robot Kinematics and Control.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Artificial Intelligence for Robotics.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Autonomous Systems and Applications.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and history of robotics; Laws of robotics; Robot classification: Industrial, Mobile, Humanoid; Robot components: Links, Joints, End effectors; Sensors (Lidar, Ultrasonic, Vision); Actuators (Servos, Stepper motors)..
2. Develop a complete answer or practical plan for: Coordinate frames and transformations; Homogeneous transformation matrices; DH parameters; Forward and Inverse kinematics (2-DOF/3-DOF); Trajectory planning; PID control for joint motion..
3. Develop a complete answer or practical plan for: Introduction to AI; Problem solving using search: BFS, DFS, A* algorithm; Expert systems; Machine Learning basics: Supervised, Unsupervised, Reinforcement learning; Neural networks and Deep learning overview..
4. Develop a complete answer or practical plan for: Path planning in dynamic environments; Obstacle avoidance; SLAM (Simultaneous Localization and Mapping) basics; Human-Robot Interaction (HRI); Industrial and service applications; Ethical issues in AI and robotics..

LAST-MILE REVISION

Quick Revision

Module 1: Robotics Fundamentals and Components

Definition and history of robotics; Laws of robotics; Robot classification: Industrial, Mobile, Humanoid; Robot components: Links, Joints, End effectors; Sensors (Lidar, Ultrasonic, Vision); Actuators (Servos, Stepper motors).

- Match each topic to CO1.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 2: Robot Kinematics and Control

Coordinate frames and transformations; Homogeneous transformation matrices; DH parameters; Forward and Inverse kinematics (2-DOF/3-DOF); Trajectory planning; PID control for joint motion.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Artificial Intelligence for Robotics

Introduction to AI; Problem solving using search: BFS, DFS, A* algorithm; Expert systems; Machine Learning basics: Supervised, Unsupervised, Reinforcement learning; Neural networks and Deep learning overview.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Autonomous Systems and Applications

Path planning in dynamic environments; Obstacle avoidance; SLAM (Simultaneous Localization and Mapping) basics; Human-Robot Interaction (HRI); Industrial and service applications; Ethical issues in AI and robotics.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Introduction to Robotics	Robotics is an interdisciplinary field involving the design, construction, and operation of robots.
Robotic Sensors and Actuators	Sensors allow robots to perceive their environment.
Kinematic Modeling	Kinematics studies the motion of robots without considering forces.
Motion Control and Planning	Trajectory planning generates a smooth path for the robot to follow.
AI Search and Problem Solving	AI enables robots to make decisions.
Machine Learning and Perception	Machine learning allows robots to improve their performance from experience.
Navigation and SLAM	Autonomous robots must navigate without human intervention.
Applications and Ethics	Robotics and AI are transforming industries.

LEARNING RESOURCES

References

Recommended robotics and AI references

1. John J. Craig, Introduction to Robotics: Mechanics and Control.
2. S. K. Saha, Introduction to Robotics.
3. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach.
4. Sebastian Thrun, Wolfram Burgard and Dieter Fox, Probabilistic Robotics.

Approved public robotics and AI sources

- https://nptel.ac.in/courses/107106090
- https://nptel.ac.in/courses/106105077

These sources provide the verified study basis while official Course 6042 detail is unavailable; they do not replace official industrial robot designs, authorised AI safety protocols, certified equipment specifications or authorised research plans.